

Art Directing Asha's Braids in Disney's "Wish"

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Figure 1: Asha Groom Creation (left-right): Sculpted Tonic tubes define the location/length of each braid at hierarchy level two. Houdini-generated braids imported into Tonic at hierarchy level three. Final groom with additional geometric detail added in Disney's XGen. Base render result with stylized geometry/materials. Final result including addition of linework and watercolor stylization.

ABSTRACT

For Walt Disney Animation Studios' 100th anniversary feature, the filmmakers wanted to honor the studio's legacy with a stylized look that draws from the rich artistic heritage of Disney's earliest films. In addition to the challenges of a highly art-directed stylized look, Asha's hairstyle comprises a full head of long, thin, tightly-braided locks which was far more complex than our previous braid glooms. In order to art direct Asha's stylized hair performance in "Wish", several advancements were made in tools and workflows across the character departments for grooming, simulation, and stylization techniques.

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1 INTRODUCTION

Asha's hairstyle was inspired by her North African heritage, and was designed as a set of long box braids which are most often posed swept to one side over her right shoulder. A series of cornrows on the left side of her scalp completes the groom. The hairstyle was strategically designed to allow for clear visibility of her face through the full range of her animation performance (see Figure 2).

There have been a number of braided hairstyles in our past films, but previously, the braids were limited to just one or two per character. Braids can be very time-consuming for artists to create, and the complexity and desired art direction of Asha's braids necessitated advancements and optimizations to our grooming and simulation tools. Her braids needed to stay on-model in all levels of her physical activity, which posed challenges for the simulation rig and shot workflows.



Figure 2: Visual development artwork

2 APPROACH

2.1 Groom Creation

For structured grooms such as Asha's, our in-house grooming tool, Tonic, is generally used to sculpt a hierarchy of tubes defining the locks of hair. To handle the level of complexity of Asha's braids, we instead leveraged a combination of Tonic and Houdini. The interactive workflows in Tonic were used to first design regions defining the overall structure of the groom as hierarchy level one. These regions are then broken out for each individual braid, and a single tube is sculpted to define the length and overall flow of the braid. The braid tube center curves are then exported to Houdini, where a procedural network is used to create the actual braided strands.

The custom Houdini braid tool is implemented as a SOP network. Controls are provided for setting the width of each strand as a percentage of the overall width, and for setting the knot frequency and the amount to vary the frequency based on the width. A ramp controls the width along the length of the braid. This network can be copied to control one or more braids with unique settings per group, then merged back together. Noise frequency is applied globally, with additional per-braid offset controls. The braid curve shapes are created using sine waves with offsets to achieve the braid knots. An additional network is provided to visualize the results while editing. Houdini's Vellum solver is used to fix intra-braid strand collisions by running the simulation while animating the braid width over a few frames. Meshes that represent the volume profile of the braid are generated by sweeping a circular profile down each braided strand. These meshes are imported back into Tonic as level three hierarchy tubes. Further refinement of the groom then continues in Tonic. The hierarchy provides control over the individual strands of a braid as well as the braid as a whole.

The resulting Tonic groom is two orders of magnitude more complex than our average grooms, containing over 500K tube vertices. Several software optimizations were done in Tonic, improving interactive rendering and selection by a factor of 60 to yield interactive rates. Significant improvements were also made to load and export times.

Our standard workflow for generating the final groom in Disney's XGen continues from this point using region maps and guide curves sampled from the Tonic tubes as input. Procedural modules add additional detail to the final full set of groom curve primitives. The overall workflow is illustrated in Figure 1.

2.2 Hair Simulation Setup

Asha's hair design calls for very specific draped shapes. Her braids form a characteristic loop that sweeps around her back and then sits in a precise shape on her shoulder. Depending on art direction and how she was framed by the camera, Asha's braids either needed to stay on her shoulder or hang fully down her back. The hair rig provided the ability to choose from four initial poses (shown in Figure 3) to help hit the art direction in motion, and to seamlessly connect to surrounding shots. The shot artists also had the ability to dial between these poses for any intermediate initial pose required. As an optimization, the rig was set up to simulate each of the 80 box braids as a single center curve with circular collision widths.

2.3 Technical Animation

The entire Technical Animation process was guided by hand-drawn sensibilities, with extensive use of 2D drawovers providing the baseline for Asha's art-directed performance. The Animation department crafted Asha's body motion, and this was supplemented with drawovers specifying the desired hair motion. An iterative collaboration between the two departments then helped define the needs of the secondary hair performance.

Staying true to the very specific draped shapes was perhaps the most difficult challenge in shots. As soon as Asha moved, the hair under physics alone would naturally fall out of the desired shape. When the camera faced her back the braids had to fall in a loop, but when the camera was viewing her from the front there needed to be less buildup of hair on her shoulder. To manage this, the artists could use the hair rig to dial in additional, shot specific shapes, depending on the camera angle and performance. Multiple simulation passes were run, which included taking hand-sculpted shapes as input to influence the results. Technical Animation continued to iterate with post-simulation hierarchical sculpting operations to achieve a final hair performance that hit the creative vision for Asha.

2.4 Hair Stylization

For the overall character stylization, a new compositing workflow was developed and packaged into a set of modifiers and mattes which could customize the hand-drawn inspired linework and the watercolor look added during rendering. Grooming techniques were modified to achieve better results with the stylization modifiers. The braids were sculpted and shaped for a graphic read, and stray hairs were reduced to keep linework clean. Breakup was added to modifier mattes to make lines less solid and give them a gestural feel. Hair silhouette edges were softened with a kuwahara filter. Linework thickness was reduced for a softer blended read between the skin and hair at the hairline. To control the shape of the braid's shadows, simplified geometry was used in lighting to define custom shadow silhouettes. Finally, to complete the watercolor stylization, new dynamic screen space texturing techniques applied watercolor and paper textures across the final rendered hairs.

3 CONCLUSION

This was a very successful collaboration across all the Character Asset and Technical Animation teams, as well as the Animation, Lighting and the hair and cloth tool developers. The updates made to our tools and workflows made it possible to meet the needs of Asha's stylized hair look and performance and moving forward will help realize future hairstyles that require braids and specific control over art-directed hair performance.

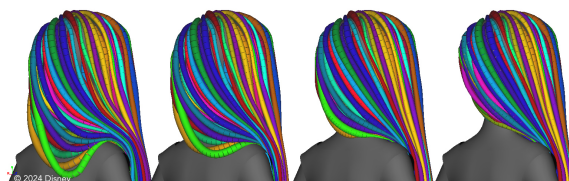


Figure 3: Hair rig initial poses for braid loop draping